📘 PROJECT GUIDE (BEGINNER FRIENDLY)

🔐 Project Title

AI-Based Password Strength Analyzer with User Authentication System

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🎯 1. What This Project Does

This system allows users to:

- 👤 Create account (with OTP verification)
- 🔐 Login (user/admin)
- 🔁 Reset password (Forgot Password + OTP)
- 🔄 Change password (after login)
- 🧠 Analyze password strength (Weak / Medium / Strong using ML)
- 🛠️ Admin can view all users

🧠 2. System Flow (VERY IMPORTANT)

```
User opens app
     ↓
Login Page
 ├── Login
 ├── Forgot Password
 └── Create Account
     ↓
If Login:
   → User Dashboard / Admin Dashboard
     ↓
User can:
   → Analyze password
   → Change password
     ↓
Admin can:
   → View all users
   → Monitor passwords
```

📂 3. Project Folder Structure

```
Password_strength/
│
├── app.py                    # Main app
├── users.json                # User data
├── model.pkl                 # ML model
├── vectorizer.pkl
│
├── utils.py                  # Password rules
├── email_utils.py            # Email OTP
│
├── training/
│   ├── train_model.py
│   ├── predict.py
│
├── dataset_making/
│   └── password_dataset.csv
```

⚙️ 4. Modules Explained (Simple Language)

🔹 app.py (Main Brain)

Handles:

- Login
- Signup
- Forgot password
- Dashboard

🔹 utils.py

Checks password rules:

```
✓ Length 8-12
✓ Uppercase
✓ Lowercase
✓ Number
✓ Special character
```

🔹 predict.py

Uses ML model to predict:

```
Weak / Medium / Strong
```

🔹 email_utils.py

Sends OTP email using Gmail SMTP

🔹 users.json

Stores:

```
{
  "user1": {
    "password": "...",
    "email": "..."
  }
}
```

🧠 5. How Password Analysis Works

🔸 Method 1: Rule-Based

Example:

- No number → weak
- No symbol → weak

🔸 Method 2: Machine Learning

- Uses dataset
- Learns patterns
- Predicts strength

🔐 6. Security Features

Feature	Description
OTP Verification	Email-based reset
Password Validation	Strong rules

Feature	Description
Role-based Login	User / Admin
ML Detection	Smart analysis

🧪 7. How to Run (Step-by-Step)

Step 1: Install libraries

```
pip install streamlit pandas scikit-learn
```

Step 2: Train model

```
cd training
python train_model.py
```

Step 3: Run app

```
cd ..
streamlit run app.py
```

👤 8. User Flow (Beginner Explanation)

🔹 Create Account

- Enter username + email
- Enter password
- System checks strength
- OTP sent to email
- Enter OTP → Account created

🔹 Login

- Enter username + password
- Redirect to dashboard

🔹 Forgot Password

- Enter username
- OTP sent
- Enter OTP
- Set new password

🔹 Change Password

- Enter old password
- Enter new password
- System validates strength

🛠️ 9. Admin Flow

- Login as admin
- View all users
- See passwords & emails
- Monitor system

🎯 10. Why This Project is Useful

💼 Real-world Applications

- Website login systems
- Banking security
- Email security systems
- Cybersecurity tools

🧠 Skills You Learn

- Python programming
- Machine Learning
- Cybersecurity basics
- Email automation
- Web app development

📊 11. Future Improvements

- 🔐 Encrypt passwords (bcrypt)
- ⏱️ OTP expiry timer
- 📊 Graphs (weak vs strong users)
- 🌐 Deploy online
- 📱 Mobile UI